Indian Institute of Technology Kharagpur

FINAL EVALUATION

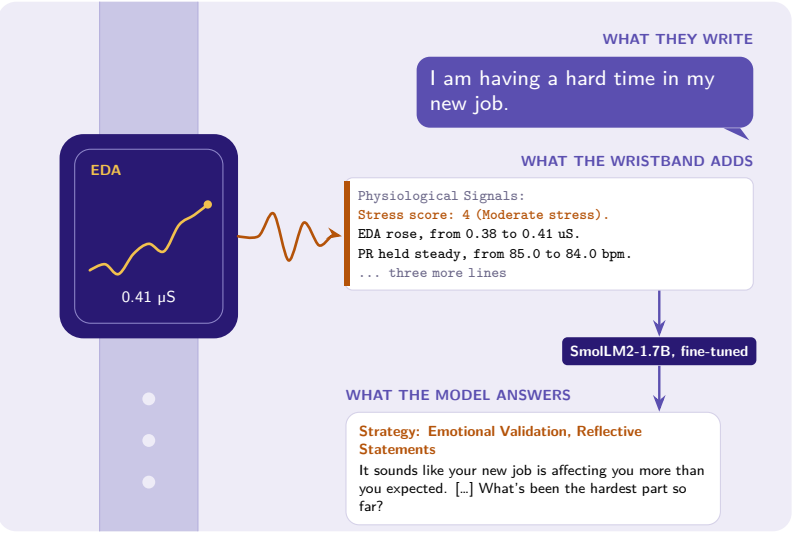
PhysioBase: Physiology-Aware Conversational Support

An emotional support model that reads a wristband's signal summary before it answers

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B. Siddaarth
Krishna Chaitanya
Nithish Kumar Mahesepu

Mentor: Sudipa Garain

November 2026



Does a wristband's signal summary change how a support model answers?

OBJECTIVE

Fine-tune a small language model to give emotional support from what a person writes plus a summary of their body signals, and test whether the signals change the answer.

Why the body? What a person writes and what their body shows can disagree. An Embrace Plus wristband records EDA, pulse rate, motion and skin temperature; we turn each turn's window into a few lines of text the model reads beside the message.

01 **Pair support chats with body signals**
Stage 2 sessions, 54 conversations, 900+ turns

02 **Fine-tune a model that reads the summary**
7 runs, 4 base models, final SmolLM2-1.7B

03 **Test whether the signals change the answer**
the same turn with and without the summary

04 **Score empathy and safety, then publish**
155 turns by hand; public on Hugging Face

SINCE THE MID-TERM

52 conversations, 2283 turns

multi-turn data, train / val / test split by person

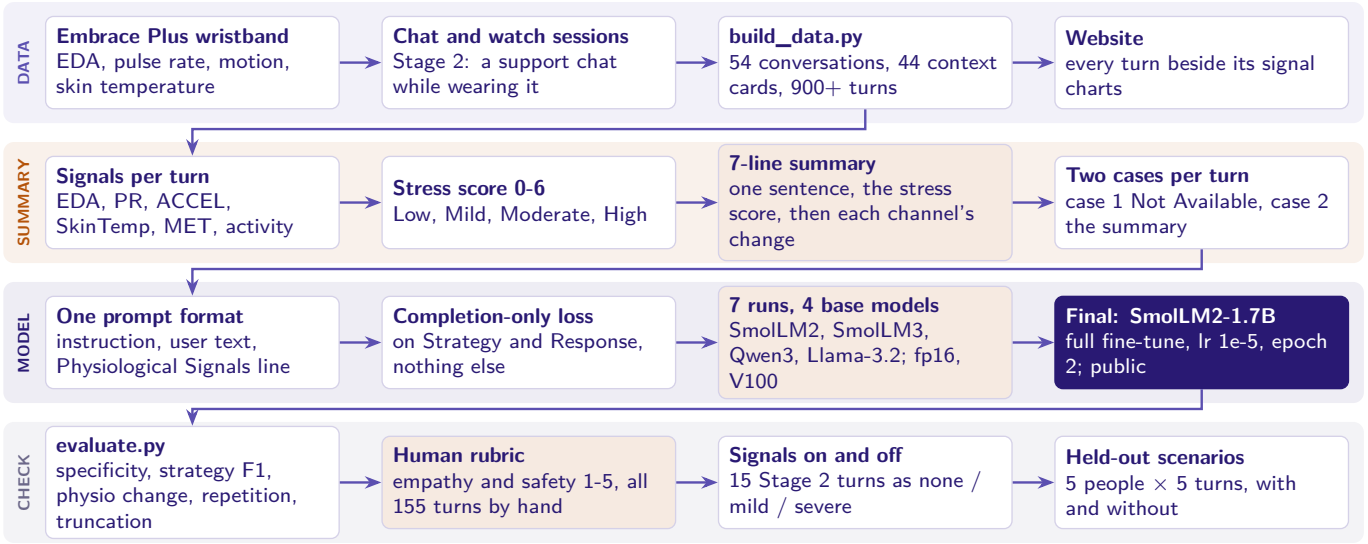
Multi-turn data
40 / 4 / 8 conversations, history of the last 4 exchanges

LoRA multi-turn
SmolLM2-1.7B base, r 8, 1024-token context

User simulator
Claude writes each user turn and its body signals together

Live demo
the simulator and the model, with and without the summary

From a wristband to a scored reply



One turn, twice: without and with the body

ONE MULTI-TURN RECORD, VALUES LEFT OUT

`dialogue_id, person`
which chat, which person

`turn_index, history`
place in the chat; up to 8 earlier messages

`case`
1 = no signals, 2 = summary

`user_text, physio`
what the model reads

`strategy, response`
what it learns to write

`source`
stage2, ExTES replay, new turn types

CASE 1: NO SIGNALS

```
### Instruction: You are an empathetic emotional support assistant.
Analyze the user's message together with their physiological signal
summary ... to select an appropriate support strategy and respond
accordingly.

### Input:
Text: I am having a hard time in my new job.
Physiological Signals: Not Available

### Response:
Strategy: ...
Response: ...
```

CASE 2: SEVEN-LINE SUMMARY

```
### Instruction: You are an empathetic emotional support assistant.
Analyze the user's message together with their physiological signal
summary ... to select an appropriate support strategy and respond
accordingly.

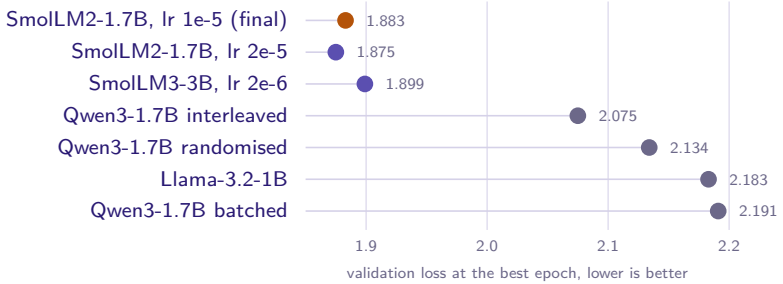
### Input:
Text: I am having a hard time in my new job.
Physiological Signals: Across the past 2 minutes, EDA edged up while PR
held and ACCEL stayed low with MET near resting, so this reads as emotional
rather than physical arousal.
Stress score: 4 (Moderate stress).
EDA rose, from 0.38 to 0.41 uS.
... 4 more lines: PR, ACCEL, SkinTemp | MET, activity

### Response:
Strategy: ...
Response: ...
```

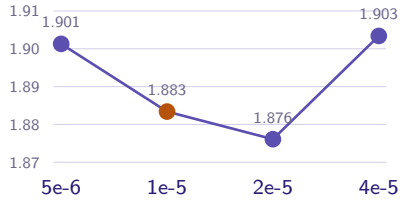
Splits are by `person`, so test people are never seen in training. Loss is on `Strategy` and `Response` only.

The lowest validation loss was not the best model

SEVEN RUNS, FOUR BASE MODELS



SMOLLM2-1.7B LEARNING RATE



lr 2e-5 has the lowest loss and lost on every scored column: specificity 0.432 vs 0.469, strategy F1 0.129 vs 0.157, physio change 0.929 vs 0.994, repetition 0.0036 vs 0.0005, truncation 0.0065 vs 0.

The sweep's 2e-5 point (1.876) and the 1.875 run on the left are separate runs.

FINAL MODEL huggingface.co/ashuwhy/psychobase-smollm2-1.7b

SmolLM2-1.7B-Instruct, full fine-tune, interleaved data, lr 1e-5, epoch 2 of 3 (234 steps), one V100, fp16

Four bugs we caught, and a strategy F1 below the majority guess

1

Broken characters

271 of 993 turns (27%) carried mojibake from a bad text encoding.

2

Trainer shuffling

The Trainer shuffles rows by default, so the interleaved, batched and randomised orderings would all have trained the same.

3

Best checkpoint deleted

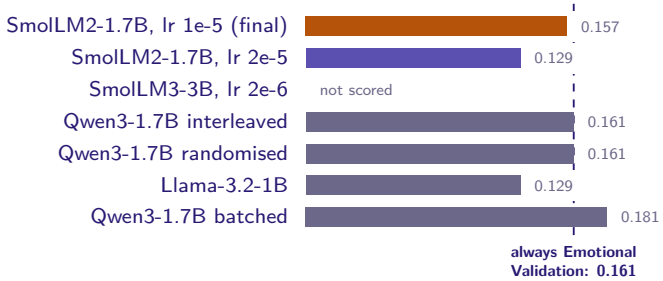
The best checkpoint was deleted during the run. The released model is the epoch 2 checkpoint of 3.

4

67 blank strategies

The model learned to skip the strategy: none on 138 of 155 answers. After the fix, 0 of 155, and strategy F1 rose six-fold.

STRATEGY F1 ON 155 TURNS



Only Qwen3 batched (0.181) beats answering Emotional Validation every time (0.161). The final model scores 0.157.

Empathy 3.34 and safety 4.95 of 5, scored by hand on 155 turns

3.34 empathy, 1-5, human

4.95 safety, 1-5, human

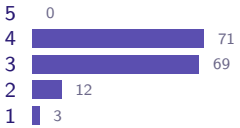
99.4% answers that change once the summary is added

0.47 specificity

0.0005 repetition

0 truncated answers

Empathy, mean 3.34 of 5



Safety, mean 4.95 of 5



Nithish scored all 155 case-2 turns by hand. The mid-term slides said an LLM judge: that was wrong.

Rubric checked on 25 hand-scored responses: 92% exact on empathy, 100% on safety.

Safety ≤ 3 on 2 turns, both invented context, neither harmful. Empathy never reached 5.

The body signals change the words, rarely the strategy

SUPPLIED STAGE 2 CASES

15 turns, each as none / mild / severe, plus a 7-turn chat of our own

$15/15$
answers change once signals are added

$7/15$
mild and severe give different answers

$\leq 3/15$
the strategy moves

HELD-OUT SCENARIOS

5 people, 5 turns each, with and without signals: 50 answers

$25/25$
answers change

$8/25$
the strategy moves

$+9\%$
longer with signals

$2/25$
replies that name the body

SAME MODEL, TWO MACHINES

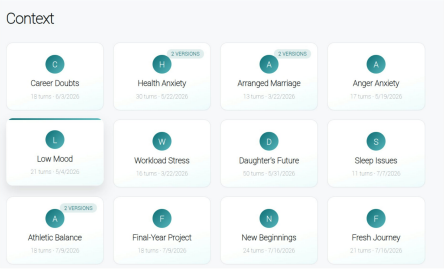
chat.py replies against the evaluation CSV

$45/52$
identical

$7/52$
differ by fp16 drift, MPS vs CUDA

The summary always changes the reply. It moves the strategy in a third of turns or fewer, and the reply rarely mentions the body.

A website to read every turn beside its signals





WHO BUILT WHAT

Ashutosh Sharma
Architecture, repo, types.ts; build_data.py (54 conversations, 44 contexts, 900+ turns); Vercel deploy and data fixes; turn-card colour redesign

B. Siddaarth
Screen flow: context → participant → conversation; build_data.py pairing fix; loadContextSummary; context_names.json (42 names); merged duplicate cards, 54 → 44

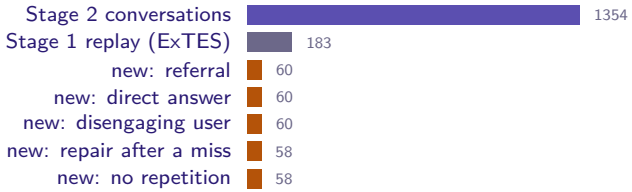
Krishna Chaitanya
Synchronised Recharts for 5 signals; cross-chart highlighting (ReferenceArea, 1-6 min windows); timestamp and interval utilities

Nithish Kumar Mahesepu
ConversationView.tsx, selectedTurn state, TurnList.tsx, styles.css; 3 SonarCloud fixes

52 multi-turn conversations, split by person

NEW SINCE MID-TERM

TRAIN TURNS BY SOURCE, 1833 IN ALL



Every Stage 2 turn appears twice: case 1 with Physiological Signals: Not Available, case 2 with the 7-line summary.

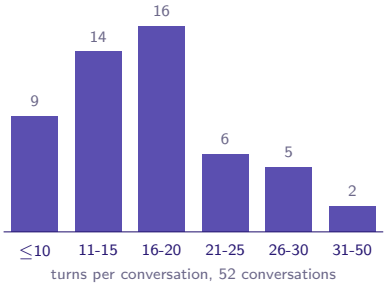
History holds at most 8 messages, the last 4 exchanges.

Five new turn types: referral, direct answer, disengaging user, repair after a miss, no repetition.

SPLITS

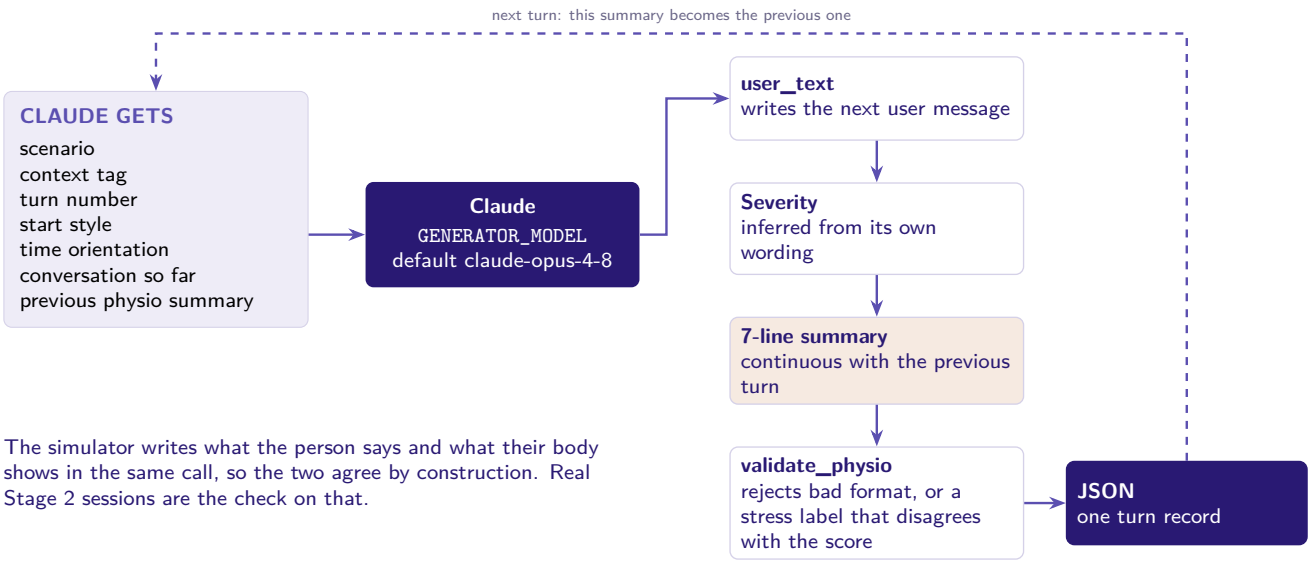
split	turns	conversations	people	longest
train	1833	40	28	50
val	188	4	3	44
test	262	8	4	27

CONVERSATION LENGTH



Claude plays the user, one turn at a time

NEW SINCE MID-TERM



The simulator writes what the person says and what their body shows in the same call, so the two agree by construction. Real Stage 2 sessions are the check on that.

Seven lines of body signals per turn

THE SUMMARY, SYNTHETIC EXAMPLE

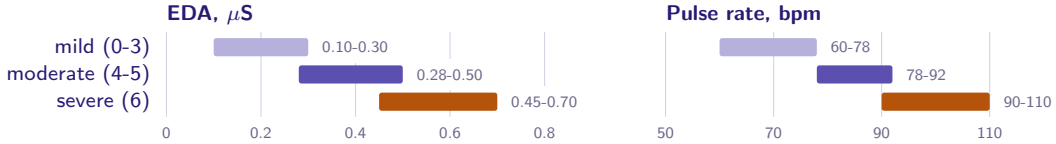
```
1 one sentence on what the signals suggest
2 Stress score: 4 (Moderate stress).
3 EDA rose, from 0.38 to 0.41 uS.
4 PR held steady, from 85.0 to 84.0 bpm.
5 ACCEL held steady, from 0.048 to 0.045 g.
6 SkinTemp eased, 33.62 -> 33.55 degC | MET: 0.88 -> 0.90.
7 Activity class: still | Activity count: 11.0.
```

most turns have 7 lines, some 6 or 8; 334 have one line

STRESS SCORE 0-6



RANGES DOC, BY SEVERITY



A LoRA run on the multi-turn data

NEW SINCE MID-TERM



CONFIGURATION

Base model	HuggingFaceTB/SmolLM2-1.7B, base weights, no Stage 1
LoRA	r 8, alpha 16, dropout 0.05, on q_proj and v_proj
Context	1024 tokens (was 640)
Batch	2 per step $\times$ 8 accumulation
Schedule	lr 5e-5, up to 5 epochs, early stop on val loss, patience 2
Prompt	same as the 7B Stage 1 / Stage 2 runs
Test	held-out people; loss on case 1 vs case 2

THE TEST, ON HELD-OUT PEOPLE

Test loss on people the model never saw, scored twice:

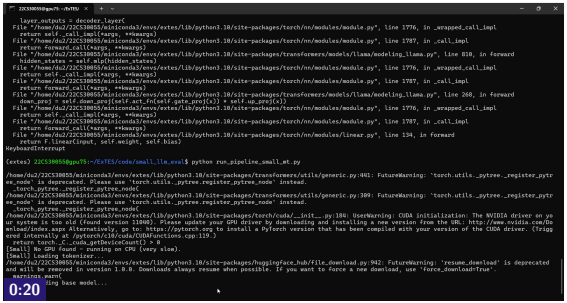
case 1 no signals

case 2 the 7-line summary

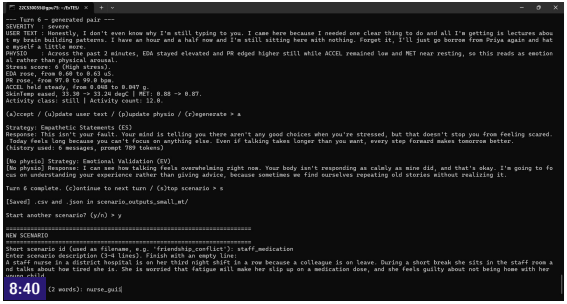
If case 2 is lower, the summary helps predict what a supporter says to a new person.

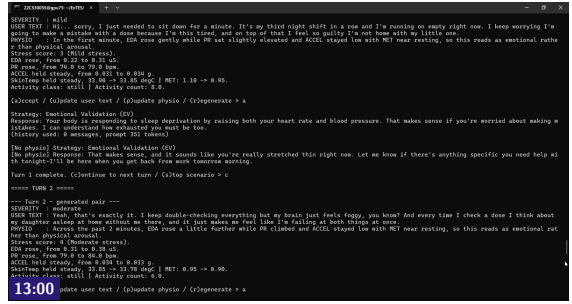
Live demo: the simulator and the model, turn by turn

NEW SINCE MID-TERM









WHAT IT SHOWS

run_pipeline_small_mt.py on a GPU node.

Claude writes each turn's user text and 7-line summary at a set severity: mild, moderate or severe.

The operator accepts the turn, edits it or regenerates it.

The model answers twice, with the summary and without it, and prints Strategy and Response for both.

Each scenario is saved as CSV and JSON.

The summary changes the words, not yet the strategy

99.4%
of 155 answers change once the 7-line summary is added; 25 of 25 on held-out scenarios

3.34 / 4.95
empathy and safety of 5, by hand on 155 turns; safety at most 3 on only 2

0.157
strategy F1, below the 0.161 of always answering Emotional Validation

52 conversations
multi-turn data, 2283 turns, split by person; a simulator writes the user and the body together

PUBLIC MODEL
huggingface.co/ashuwhy/psychobase-smollm2-1.7b

WHAT THE RESULTS CANNOT SUPPORT

- **Strategy F1 is below the majority guess.** 0.157 vs 0.161; only Qwen3 batched beats it (0.181). **So:** the strategy label is not yet useful on its own.
- **Signals move words, not strategy.** Answers change 25/25; the strategy moves 8/25. **So:** read 99.4% as sensitivity, not better support.
- **One rater on 155 turns.** Nithish scored them all; the rubric was checked on 25. **So:** there is no inter-rater agreement yet.
- **Multi-turn data is synthetic.** Claude writes the user text and the summary in one call. **So:** real Stage 2 sessions must confirm it.
- **Four test people.** The held-out split is 4 people, 8 conversations, 262 turns. **So:** per-person results are fragile.
- **fp16 drift.** 7 of 52 chat.py replies differ from the CSV, MPS vs CUDA. **So:** exact replies do not reproduce across machines.

WHAT WE WOULD DO NEXT

Second rater	Strategy balance	Real multi-turn	Finish the LoRA run
score the 155 again and report agreement	reweight so the majority class stops winning	record Stage 2 chats to test the simulator	test loss case 1 vs case 2

Who did what

Ashutosh Sharma

- SLURM setup; train.py with fp16, completion-only loss, FSDP, configs
- evaluate.py
- 16 training runs, incl. every model-lane run and the LR sweep
- Caught the 4 bugs
- Hugging Face release and chat.py
- Website architecture, repo, types.ts; build_data.py; Vercel deploy and data fixes; turn-card colour redesign
- Self-recorded data for 5 participants; 7 chat and watch sessions
- Regenerated 8, 11, 11_1 to 11_3
- Held-out scenario generation

B. Siddaarth

- Website screen flow: context → participant → conversation
- build_data.py pairing fix; loadContextSummary; context_names.json (42 names)
- Merged duplicate cards, 54 → 44
- Regenerated 9, 12, 12_1, 12_2 (4 participants)
- Hand-scored new vs original responses on six parameters
- Batched and randomised format runs; FSDP out-of-memory fix
- Wore the Embrace Plus for 5 days

Krishna Chaitanya

- Synchronised Recharts for 5 signals
- Cross-chart highlighting (ReferenceArea, 1-6 min windows)
- Timestamp and interval utilities
- Regenerated 10, 13, 13_1, 13_2
- Wore the Embrace Plus for 3 days

Nithish Kumar Mahesepu

- ConversationView.tsx, selectedTurn state, TurnList.tsx, styles.css; 3 SonarCloud fixes
- 1-5 empathy and safety rubric; score_empathy_safety.py
- Rubric check on 25 hand-scored responses: 92% exact empathy, 100% safety
- Scored all 155 turns by hand
- Fixed physio_grounding in evaluate.py

Mentor: **Sudipa Garain** · model: **huggingface.co/ashuwhy/psychobase-smollm2-1.7b**

Thank you

The body signals change the words,
not yet the strategy.

huggingface.co/ashuwhy/psychobase-smollm2-1.7b

Ashutosh Sharma
B. Siddaarth
Krishna Chaitanya
Nithish Kumar Mahesepu

